

## HARSTAD, Norway

WMO: 011800

Lat: 68.802N

Lon: 16.537E

Elev: 148

StdP: 14.62

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 13.6       | 15.8 | 2.3                           | 6.2 | 16.3 | 4.7 | 7.0 | 17.8 | 21.0                  | 28.5 | 17.5 | 28.7 | 3.8                   | 210  | 0.647 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.3                   | 71.1            | 59.1 | 67.8 | 58.0 | 64.5 | 56.5 | 61.2                | 68.9 | 59.0 | 66.0 | 57.2 | 63.1 | 3.9                  | 40   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 57.4                            | 70.6 | 65.0 | 55.5 | 66.0 | 63.0 | 53.7 | 61.7 | 60.8 | 27.3          | 69.1 | 25.8 | 66.3 | 24.6 | 63.2 | 70.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |      |      |     |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|------|------|-----|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p)  |      |     |
| (4) 15.8          | 12.9 | 11.0 | DB                         | 9.5 | 77.2               | 3.5 | 5.7                                                | 6.9 | 81.3       | 4.9 | 84.6       | 2.9 | 87.8       | 0.4  | 92.0 | (4) |
| (5)               |      |      | WB                         | 7.8 | 63.2               | 3.2 | 3.2                                                | 5.5 | 65.5       | 3.7 | 67.4       | 1.9 | 69.2       | -0.4 | 71.5 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 40.4  | 27.8  | 27.7  | 30.5  | 36.6  | 44.6  | 49.9  | 56.1  | 54.7  | 48.8  | 40.6  | 34.3 | 31.9 | (6)  |
| (7)  |                                                                   | DBStd       | 11.44 | 6.50  | 6.80  | 6.10  | 5.36  | 5.97  | 5.40  | 5.51  | 4.80  | 5.10  | 5.95  | 5.65 | 5.75 | (7)  |
| (8)  |                                                                   | HDD50       | 3995  | 688   | 623   | 606   | 402   | 188   | 66    | 5     | 10    | 82    | 295   | 470  | 560  | (8)  |
| (9)  |                                                                   | HDD65       | 8993  | 1153  | 1043  | 1071  | 852   | 632   | 454   | 282   | 319   | 487   | 755   | 920  | 1025 | (9)  |
| (10) |                                                                   | CDD50       | 486   | 0     | 0     | 0     | 1     | 21    | 62    | 195   | 157   | 45    | 5     | 0    | 0    | (10) |
| (11) |                                                                   | CDD65       | 9     | 0     | 0     | 0     | 0     | 0     | 1     | 7     | 1     | 0     | 0     | 0    | 0    | (11) |
| (12) |                                                                   | CDH74       | 61    | 0     | 0     | 0     | 0     | 0     | 4     | 50    | 7     | 0     | 0     | 0    | 0    | (12) |
| (13) |                                                                   | CDH80       | 11    | 0     | 0     | 0     | 0     | 0     | 0     | 9     | 2     | 0     | 0     | 0    | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 5.1   | 5.5   | 5.5   | 5.7   | 5.1   | 5.3   | 5.3   | 4.6   | 4.3   | 4.2   | 4.9   | 5.0  | 5.4  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 39.1  | 3.9   | 3.9   | 3.8   | 2.4   | 2.1   | 1.8   | 2.4   | 2.7   | 3.7   | 4.6   | 3.5  | 4.7  | (15) |
| (16) |                                                                   | PrecMax     | 46.4  | 8.0   | 7.8   | 8.0   | 5.0   | 3.7   | 3.0   | 5.1   | 6.1   | 7.7   | 7.2   | 7.3  | 10.6 | (16) |
| (17) |                                                                   | PrecMin     | 30.4  | 0.7   | 0.7   | 1.1   | 0.3   | 0.6   | 0.3   | 0.6   | 1.3   | 0.9   | 1.2   | 0.0  | 0.3  | (17) |
| (18) |                                                                   | PrecStd     | 4.7   | 2.0   | 1.7   | 1.9   | 1.2   | 0.9   | 0.7   | 1.3   | 1.1   | 1.8   | 1.7   | 2.1  | 2.7  | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 43.7  | 44.1  | 47.4  | 55.8  | 68.8  | 72.4  | 80.1  | 72.7  | 64.3  | 57.6  | 50.8 | 46.5 | (19) |
| (20) |                                                                   |             | MCWB  | 39.0  | 40.1  | 38.9  | 44.8  | 53.3  | 57.6  | 65.4  | 60.4  | 54.1  | 49.6  | 46.1 | 42.2 | (20) |
| (21) |                                                                   | 2%          | DB    | 40.9  | 41.5  | 43.3  | 51.3  | 63.0  | 66.7  | 72.8  | 68.6  | 61.2  | 54.1  | 46.8 | 44.0 | (21) |
| (22) |                                                                   |             | MCWB  | 37.4  | 37.6  | 37.9  | 43.2  | 51.5  | 55.3  | 60.7  | 59.4  | 53.7  | 47.7  | 42.8 | 40.3 | (22) |
| (23) |                                                                   | 5%          | DB    | 38.6  | 39.6  | 40.9  | 47.0  | 57.6  | 62.0  | 69.1  | 65.4  | 58.5  | 51.7  | 44.0 | 41.5 | (23) |
| (24) |                                                                   |             | MCWB  | 35.5  | 36.3  | 36.4  | 40.8  | 48.1  | 53.1  | 59.5  | 57.8  | 52.4  | 47.0  | 40.1 | 37.7 | (24) |
| (25) |                                                                   | 10%         | DB    | 36.6  | 37.3  | 39.0  | 44.4  | 53.7  | 58.8  | 65.8  | 62.5  | 56.2  | 49.3  | 41.9 | 39.5 | (25) |
| (26) |                                                                   |             | MCWB  | 33.5  | 34.4  | 35.2  | 39.1  | 45.9  | 51.5  | 57.8  | 56.0  | 51.3  | 45.2  | 38.6 | 36.0 | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 39.6  | 40.6  | 40.2  | 45.8  | 54.9  | 59.2  | 65.9  | 62.8  | 56.4  | 50.4  | 46.7 | 42.7 | (27) |
| (28) |                                                                   |             | MCDWB | 42.3  | 43.7  | 44.8  | 54.0  | 66.0  | 69.8  | 77.7  | 68.5  | 61.7  | 55.6  | 49.8 | 45.4 | (28) |
| (29) |                                                                   | 2%          | WB    | 37.8  | 38.2  | 38.6  | 43.6  | 51.7  | 56.4  | 62.5  | 60.3  | 54.7  | 49.0  | 43.4 | 40.8 | (29) |
| (30) |                                                                   |             | MCDWB | 40.4  | 40.9  | 42.5  | 50.1  | 61.9  | 64.5  | 71.5  | 67.2  | 59.4  | 53.3  | 46.2 | 43.6 | (30) |
| (31) |                                                                   | 5%          | WB    | 35.8  | 36.6  | 37.1  | 41.4  | 48.9  | 54.0  | 60.2  | 58.4  | 53.1  | 47.3  | 40.7 | 38.3 | (31) |
| (32) |                                                                   |             | MCDWB | 38.3  | 39.2  | 40.4  | 46.4  | 57.0  | 61.0  | 67.5  | 64.1  | 57.4  | 51.1  | 43.1 | 41.0 | (32) |
| (33) |                                                                   | 10%         | WB    | 33.9  | 34.7  | 35.5  | 39.5  | 46.6  | 51.9  | 58.2  | 56.6  | 51.6  | 45.4  | 38.8 | 36.3 | (33) |
| (34) |                                                                   |             | MCDWB | 36.2  | 37.1  | 38.7  | 43.9  | 53.0  | 58.1  | 64.8  | 61.7  | 55.9  | 49.0  | 41.5 | 38.9 | (34) |
| (35) | Mean Daily Temperature Range                                      | 5% DB       | MDBR  | 7.1   | 7.6   | 8.9   | 10.0  | 10.9  | 10.2  | 11.3  | 10.5  | 9.6   | 7.4   | 6.6  | 7.0  | (35) |
| (36) |                                                                   |             | MCDBR | 8.0   | 8.7   | 8.7   | 12.5  | 17.1  | 16.0  | 17.6  | 15.0  | 12.1  | 8.8   | 7.8  | 8.7  | (36) |
| (37) |                                                                   |             | MCWBR | 6.8   | 7.2   | 6.2   | 7.9   | 9.3   | 8.8   | 8.8   | 8.0   | 7.3   | 6.3   | 6.1  | 7.3  | (37) |
| (38) |                                                                   | 5% WB       | MCDWB | 7.6   | 8.5   | 8.6   | 11.9  | 16.5  | 14.9  | 16.6  | 13.7  | 11.0  | 8.7   | 7.4  | 8.7  | (38) |
| (39) |                                                                   |             | MCWBR | 6.9   | 7.3   | 6.5   | 8.1   | 9.3   | 8.8   | 8.7   | 7.9   | 7.1   | 6.8   | 6.2  | 7.7  | (39) |
|      |                                                                   |             |       |       |       |       |       |       |       |       |       |       |       |      |      |      |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.372 | 0.268 | 0.308 | 0.329 | 0.339 | 0.340 | 0.344 | 0.334 | 0.317 | 0.278 | 0.351 | N/A  | (40) |      |
| (41) |                                                                   | taud        | 2.653 | 2.116 | 2.209 | 2.326 | 2.444 | 2.498 | 2.515 | 2.531 | 2.542 | 2.261 | 2.620 | N/A  | (41) |      |
| (42) |                                                                   | Ebn at Noon | 30    | 156   | 224   | 258   | 270   | 273   | 268   | 258   | 231   | 153   | 25    | 0    | (42) |      |
| (43) |                                                                   | Edh at Noon | 3     | 14    | 24    | 29    | 29    | 28    | 27    | 24    | 18    | 12    | 3     | 0    | (43) |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 6     | 95    | 430   | 999   | 1497  | 1553  | 1400  | 999   | 532   | 176   | 20    | 0    | (44) |      |
| (45) |                                                                   | RadStd      | 1     | 20    | 69    | 90    | 121   | 209   | 147   | 91    | 56    | 31    | 2     | 0    | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (25 neighbors) | +0.61 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page